Complete Requirements Table

SE Lab-1 | Problem Statement #48: Incident Escalation & On-Call Rotation Engine

Name: Aryan N SRN: PES1UG24CSS656

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-001	Functional	The system shall ingest monitoring alerts and identify the active on-call engineer from the rotation schedule.	High	**PASS:** The alert is assigned to the active on-call engineer. **FAIL:** The alert is assigned to an unavailable engineer.	Ensures alerts are immediately routed to the correct active engineer.
FR-002	Functional	The system shall escalate an incident to the secondary on-call engineer if it is not acknowledged within 5 minutes.	High	**PASS:** An unacknowledged incident is automatically escalated after 5 minutes. **FAIL:** The incident remains with the primary engineer without escalation.	Prevents critical incidents from remaining unattended.

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Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-003	Functional	The system shall notify the assigned on-call engineer through configured phone/SMS escalation channels.	High	**PASS:** The engineer is successfully notified through the configured channel. **FAIL:** The system fails to send the notification through the required channel.	Provides rapid communication so engineers can respond quickly.
FR-004	Functional	The system shall allow the Incident Commander to track incident status and acknowledgements.	Medium	**PASS:** The Incident Commander can view the required incident details, status, and acknowledgements. **FAIL:** Required incident information is unavailable.	Enables effective monitoring and coordination during incidents.

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Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-005	Functional	The system shall display and maintain post-mortem information, including incident details, actions taken, escalation events, and resolution.	Medium	**PASS:** A complete post-mortem can be recorded and retrieved. **FAIL:** Required incident or resolution information cannot be stored or retrieved.	Supports incident analysis and improves future incident response.

Complete Requirements Table

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Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
NFR-001	Nonfunctional (Performance and Security)	Alert dispatch through webhook, SMS, and email shall be initiated within 3 seconds of alert ingestion under expected load.	High	**PASS:** Benchmark testing confirms that alert dispatch is initiated within 3 seconds. **FAIL:** Dispatch exceeds the 3-second target under simulated peak load.	Rapid alert delivery is essential for timely incident response.
NFR-002	Nonfunctional (Reliability and Availability)	The system shall maintain reliable alert routing and escalation without losing incidents or escalation events during normal and simulated peak operation.	High	**PASS:** All test alerts and escalation events are successfully recorded and processed. **FAIL:** Any alert or escalation event is lost or incorrectly processed.	Prevents missed incidents and ensures dependable system operation.